

RESEARCH STATEMENT

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My research interests lie at the intersection of **computational and applied mathematics**, with a current focus on developing innovative analytical tools and crafting high-order accurate and stable numerical methods. These tools and methods find applications across a wide spectrum of disciplines. My research aspirations lie in **harnessing advanced mathematics to unveil novel insights on consequential real-world challenges** – seeding fresh perspectives that deepen scientific comprehension while pioneering qualitative and quantitative tools for incisive prediction.

My doctoral work focused on developing rigorous numerical methods to study the existence and properties of smooth invariant sets in dynamical systems. I studied the role of invariant objects like stable/unstable manifolds in organizing the structure of phase space. This work uses deliberate numerical simulations involving high order calculations to suggest conjectures about global dynamics, with a special emphasis on the appearance and subsequent destruction of invariant tori in dissipative systems. These computational methods are amenable to a-posteriori analysis, so that many of the conjectures can be verified using constructive methods of computer-assisted proof. I seek to continue exploring serious scientific questions by **harnessing numerical analysis, data assimilation and applied dynamical systems toolkits** – spanning interests from spectral stability to the theory (and numerical tools) behind critical transitions.

I will now summarize the major accomplishments and future directions of my research. History and references to related work are mostly omitted and can be found in the cited publications.

Understanding Tipping Phenomena by Computing Most Probable Escape Paths (MPEPs)

Rate-induced tipping (R-tipping) occurs when a ramp parameter changes rapidly enough to cause the system to tip between co-existing, attracting states, while **noise-induced tipping (N-tipping)** occurs when there are random transitions between two attractors of the underlying deterministic system. A significant portion of my recent work has been dedicated to **identifying the threshold for tipping**. In the canonical case, the basin boundary is defined by the stable manifold (W^s) of a fixed point. However, basin boundaries can exhibit far more intricate structures, even in systems with seemingly straightforward bifurcation structures of basic orbits, such as fixed points and periodic orbits^{2;4;23;25}. The crux of the challenge lies in **computing and confidently determining the intricate structure of these basin boundaries**.

In the realm of N-tipping, recent findings by Ritchie and Sieber²⁴ have unveiled that the presence of noise can compel tipping to occur at a subcritical rate. The extent to which this connection between the heteroclinic orbit and the Most Probable Escape Path (MPEP), a concept from Large Deviation Theory¹⁸, holds in a general sense remains an open question. When the basin boundary (tipping threshold) becomes intricate, such as the stable manifold of a periodic orbit¹⁷ or a chaotic set itself^{9;22}, the factors determining the MPEP are not well understood. The works of E and Zhou³¹ and Cameron⁷ have highlighted the challenges and resolved many cases, but the majority of research on MPEPs has operated under the assumption that the underlying dynamics possess a certain gradient structure. I have made some pertinent contributions in that direction²⁸. In my work, I showed that there is a distinguished MPEP crossing a periodic orbit basin boundary which acts as a guide for noisy escaping paths in the case of small noise slightly away from the limit of vanishing noise (see Figure 1).

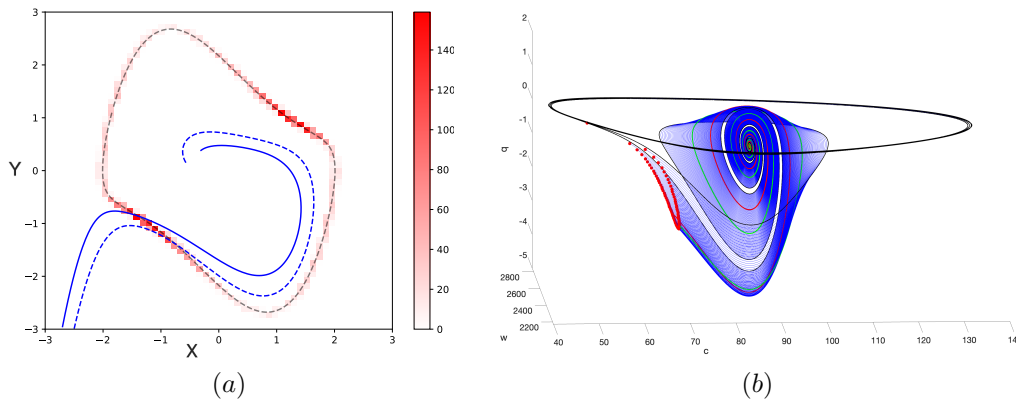


Figure 1: In (a) the solid blue curve marks the predicted most probable escape path as determined by minimizing the action of the OM functional for the inverted Van der Pol equations. The dashed blue curve depicts the trajectory that corresponds to the pivot point (boundary for local minimizers). The heat map displayed on the periodic orbit depicts the probability of an escape event occurring at that location. The red curve in Figure (b) represents the MPEP, and the green curve, the pivot point for the carbon cycle model. The red dots are where trajectories that escape intersect the torus on the basin boundary. The blue trajectories are *River* trajectories delineated by heteroclinic orbits (black).

In another on-going project, I study rate-induced tipping in reaction-diffusion equations (RDEs) in one spatial dimension, where the reaction term decays in space (asymptotically homogeneous) and varies linearly with time (nonautonomous)

due to an external input. I introduce a time-dependent advection term so that the system will have the form:

$$u_t = Du_{zz} + c(t)u_z + f(u, H(z)),$$

$$\lim_{z \rightarrow \pm\infty} u(z, t) = 0$$

where $c(t)$ represents the speed of habitat change. The reaction term is also non-autonomous and represents a spatially localized, single-species, favorable habitat zone moving from one asymptotic location to another. I then parameterize a nonlinear ramp function

$$c(t) = \gamma(rt, d), \quad (1)$$

so that r quantifies the rate of change of the environment for a fixed displacement d . The movement is thus characterized by the displacement d and the rate parameter r , which quantifies the time scale on which it occurs. In this work, I identify a critical displacement d^* and, for $d > d^*$, demonstrate the existence of a critical rate $r^*(d)$ at which the system undergoes rate-induced tipping: for $r > r^*$ an initially thriving habitat population becomes extinct due to too rapid habitat change, see Figure 2.

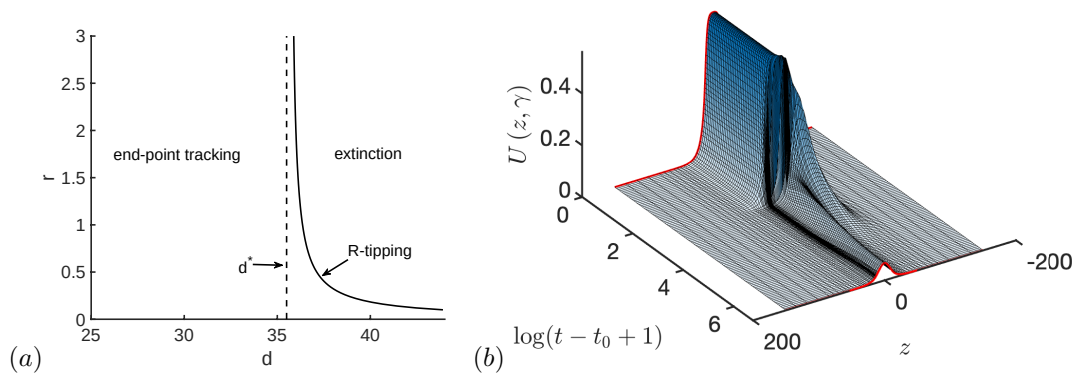


Figure 2: (a) Plot of the critical value r^* against d for a fixed habitat length L . The dashed vertical line approximates the location of the critical displacement d^* . (b) For $r = r_*$, the system undergoes R-tipping. We thus have a heteroclinic connection from the stable pulse (red) at $t = -\infty$ to the unstable pulse (red) at $t = \infty$, representing the threshold for R-tipping.

Computational Dynamical Systems

The next part of my work focuses on **building the computational technology needed to study the geometry behind the dynamics of these complex systems**. This work comprised a detailed numerical study of mostly dissipative vector fields. The purposeful calculations and three-dimensional renderings of invariant manifolds throughout my works furnished more delicate insights into the dynamics of the system in question than are obtained by straightforward simulations of ensembles of initial conditions^{5;13}.

In an ongoing project⁶, an implicit real analytic computational method for studying billiard maps on tables that are small perturbations of ellipses is developed (Figure 3a), enabling the calculation of stable and unstable manifolds (Figure 3b). Dynamical billiards involve a particle moving freely on a two-dimensional table and bouncing elastically off the boundary, creating a sequence of collision points and reflection angles that form a discrete iterated map first analyzed by Birkhoff. Although elliptical tables exhibit complete integrability - meaning their dynamics can be solved exactly — the renowned Birkhoff conjecture proposes that ellipses are the unique smooth convex shapes with this property. Though the current results are numerical, the method is designed with sufficient mathematical rigor to potentially support future computer-assisted proofs and validation.

In another ongoing project, I embark on a journey to **unravel the intricate driving mechanisms that govern the stability of travelling waves**. Stability analysis of waves or coherent structures often hinges on the resolution of an eigenvalue problem, which emerges from the linearization of the governing equations around the steady state representing the wave or coherent structure. This fundamental framework has been generalized and extended in various ways to study stability for specific applications. I aim to **prove stability of these coherent structures both analytically and numerically** using Sturm-Liouville theory and its generalization **using the Maslov Index**. Pioneering work in this area was conducted by Chris Jones^{20;21}, followed by collaborative efforts with other researchers^{3;8;19}.

My analysis combines dynamical systems techniques with a separation of scales argument¹⁵, providing a novel framework for studying spectral properties of differential operators where multiple spatial scales interact, see Figure 4. A current study in this line of work is the nonlinear Schrödinger (NLS) equations with an added potential¹⁴. I take the potential as radial and having a particular scaling and set $V_\epsilon(|x|) = \epsilon^2 V(\epsilon|x|)$ where V may be any fixed potential that I will take to

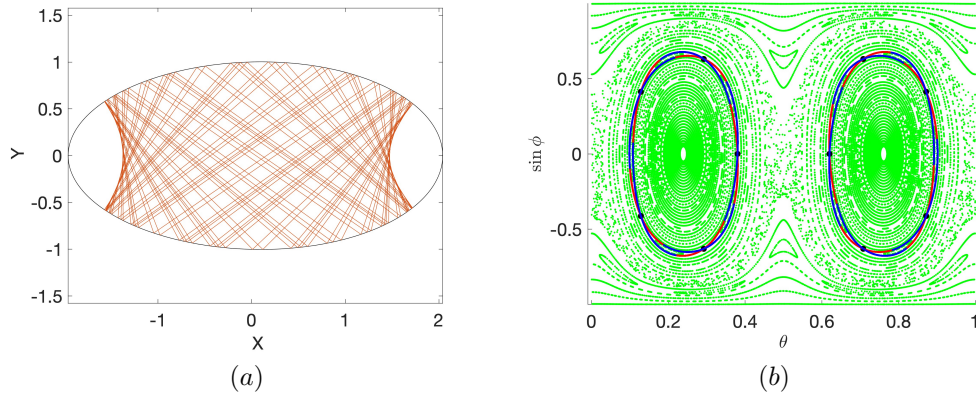


Figure 3: (a) Physical trajectory showing a particle's path (orange) bouncing within the deformed elliptical boundary (black). (b) Phase space representation displaying the computed stable manifold (blue) and unstable manifold (red) associated with a period-10 periodic orbit, calculated using the implicit real analytic parametrization method. The dense green structure reveals the complex dynamics that emerge when the elliptical boundary is perturbed, showing intricate phase space behavior.

be exponentially decaying at ∞ . The full evolution equation is given by:

$$i \frac{\partial \varphi}{\partial t} = \Delta \varphi + |\varphi|^2 \varphi - V_\epsilon(|x|) \varphi. \quad (2)$$

I work in $\mathbb{R}^3$ so that $x = (x_1, x_2, x_3)$ and $\Delta = \sum_{i=1}^3 \frac{\partial^2}{\partial x_i^2}$. Note that I can think of (2) as an evolution equation on some function space X of complex-valued functions φ , for instance: $X = L_2(\mathbb{R}^3)$. Because $\mathcal{L} = JL$ is Hamiltonian, its spectrum has four-fold symmetry: if λ is in the spectrum so are $\bar{\lambda}$, $-\lambda$, and $-\bar{\lambda}$. The essential spectrum lies entirely on the imaginary axis. There may be spectrum off the imaginary axis, in which case there is inevitably unstable spectrum. If the spectrum is entirely on the imaginary axis, then there is the potential for stability, but not a guarantee. The ultimate objective of this project can be articulated as follows: I aim to demonstrate that, in the absence of any spectrum off the imaginary axis, the standing wave under consideration exhibits stability. This notion of stability is understood in the ‘‘Hamiltonian’’ sense, implying that initial conditions in close proximity to the standing wave will give rise to solutions that remain bounded and do not diverge over time. If there are no eigenvalues present in the spectral gap, then I can rigorously establish stability in this context. However, I anticipate the emergence of gap eigenvalues when $\epsilon > 0$, and the challenge then lies in accurately enumerating them.

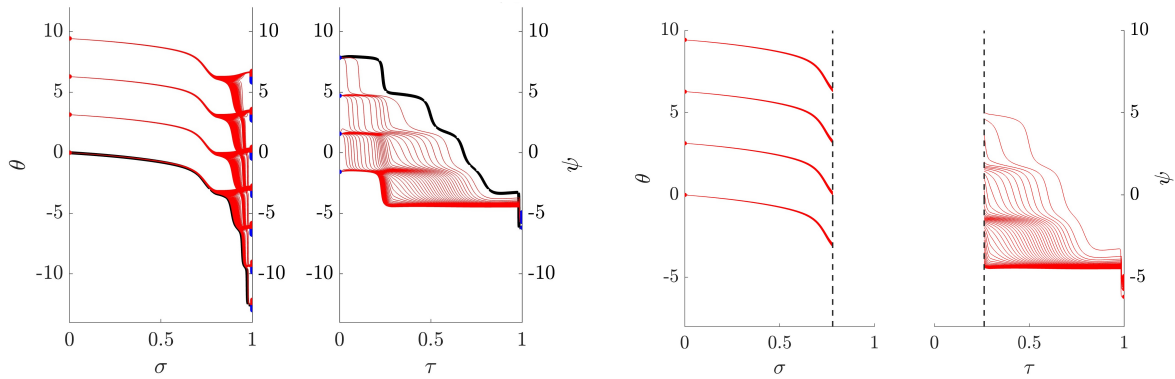


Figure 4: Using $V_0(x) = \frac{-2.6}{1+2x^4}$, $V_{1,\epsilon}(x) = -20\epsilon^2 \text{sech}(\epsilon x)$, we track the evolution of trajectories from two systems, (θ, σ) and (ψ, τ) interacting at different spatial scales up to their respective threshold values where topological gluing occurs (Left). The image on the right shows the maximum jumps in multiples of π in black, giving the full eigenvalue count of $\Delta - V_0 - V_{1,\epsilon}$.

Motivated by problems in ecology, I also study standing wave solutions of a reaction-diffusion equation modeling population dynamics and their stability. I use the following nonlinear reaction-diffusion equation (RDE) in one spatial dimension to do so:

$$u_t = Du_{xx} + f(u, H(x)),$$

with an Allee growth term

$$f(u, H(x)) = -\beta u + \lambda H(x)u^2 - \gamma u^3$$

where

$$H(x) = \begin{cases} h^* \in \mathbb{R}, & \text{if } |x| \leq \frac{L}{2} \\ 0, & \text{otherwise} \end{cases}$$

with constant diffusion coefficient $D > 0$ and boundary conditions

$$\lim_{x \rightarrow \pm\infty} u(x, t) = 0.$$

Central to this study is the existence of a finite favorable habitat patch where positive growth rates for the species are attained with the remainder of the spatial domain unfavorable to species existence. Using dynamical systems techniques, I study the existence of pulse-like standing waves and how these solutions depend on system parameters for an Allee type growth model²⁶, see Figure 5. I also plan to use Sturm-Liouville theory and topological methods to rigorously validate the stability of these pulse-like solutions.

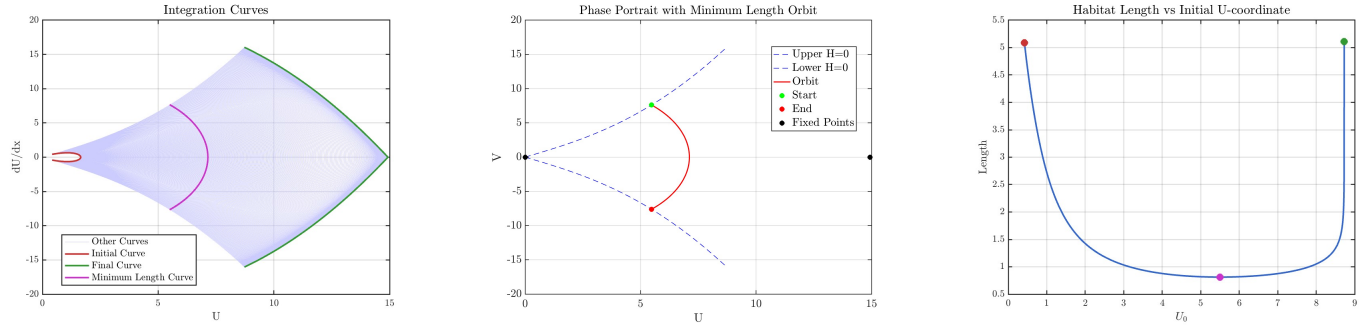


Figure 5: (a) Phase space trajectories for standing wave solutions of the RDE; (b) Phase portrait of the minimum length orbit (red) representing the critical patch size for existence; (c) Habitat length function vs initial position U_0 for the RDE which guarantees the existence of a critical length $L^* > 0$ such that if $L > L^*$, the habitat has exactly two steady-state pulse solutions; otherwise, there are zero for $L < L^*$.

Data Assimilation

I have experience in **using an iterative ensemble smoothing method called an ensemble smoother with multiple data assimilation (ESMDA)**^{10–12;30}. Concrete projects that I am involved in study climate models which allow for good data fitting and creating new algorithms altogether²⁷, see Figure 6a. I was also in a project involving disease transmission among sub-populations¹⁶, see Figure 6b. In another research project, I investigated ocean circulation changes through the lens of data assimilation using a reduced-order model with undergraduate and graduate students from different institutions around the United States of America²⁹.

I plan to **use Most Probable Escape Paths to inform ensemble initialization and evolution**. For a system with state $x \in \mathbb{R}^n$ and observations y , the ensemble Kalman filter update is: $x_i^a = x_i^f + K(y - H(x_i^f))$, where K is the Kalman gain matrix and H is the observation operator. Near critical transitions, this update can fail because: i) the ensemble spread doesn't capture important transition pathways, ii) the linear observation operators miss nonlinear relationships, and iii) the Gaussian assumptions break down. By combining MPEP computations, I can design ensemble schemes that effectively capture and predict such events. This approach can be seen as a novel alternative to the standard method of inflation used when a filter collapses due to an attracting state.

Future Work

Tipping phenomena in dynamical systems. My first problems in this context will be built around existing work. The interactions in tipping phenomena remains largely untapped for applications. Using previously built algorithms, I intend to study phase-based, data-driven models in collaboration with scientists in the field. I also aim to study complex basin boundaries in 3-dimensional systems generalizing the 2-dimensional models we tackled. For stochastic dynamics, near-term plans include both theory and algorithm development. In addition to the foundations, I intend to explore questions such as whether mean-transition-time and mean-first-exit-time can play a useful role in transitions from (meta)stable states. The noise-facilitated rate-induced tipping phenomenon can be regarded as a unique case of noise-induced tipping, characterized by the absence of noise in the equation governing the rate parameter. This area of research remains largely unexplored, presenting a wide array of open questions and opportunities for further investigation. The degeneracy inherent in this situation sets it apart from the well-studied gradient case, rendering the applicability of the MPEP theory uncertain. A fundamental challenge emerges from the non-gradient structure of the underlying equations, which gives rise to the possibility of intricate basin boundaries, often referred to as tipping thresholds¹⁸. I plan to extend work in this direction

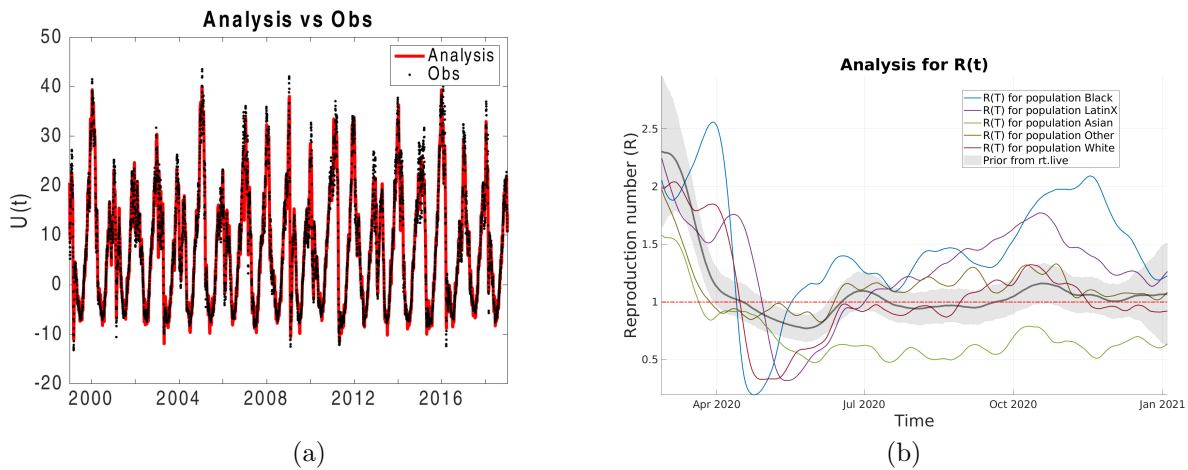


Figure 6: In (a), we see a comparison of model estimated wind speeds (red) using ESMDA with the observed wind speeds (black) collected from ECMWF reanalysis averaged data for the Ruizmaikin model. (b) Analysis $R_t(n)$ results for the continuous update case for the District of Columbia which reveals that the the Black community (blue) was mostly affected by the spread of SARS-CoV-2.

as well.

Mathematical analysis and computational methods for applied problems. I plan to continue my research on mathematical modeling to gain further insights into many important natural phenomena in the physical sciences at different scales. My research in this direction will contribute to the understanding of a variety of systems in biology and ecology, among others. This leads to developing appropriate descriptive measures to quantify the spatiotemporal and structural complexity of physical systems and then to build up analytically tractable mathematical models to capture intrinsic dynamics. Computer-assisted proof methods are largely missing in the tipping literature, so I plan to develop computational tools for that as well. For further development, I will guide students and join collaborators to theoretically and numerically investigate several separate but ultimately related problems at the intersection of applied and computational mathematics. This collection of problems concentrates on the development of mathematical tools and my main purpose is to provide a meaningful description and novel treatment of systems that arise from physics, biology, and other natural and social sciences.

Data assimilation schemes for non-Gaussian dynamics near tipping points. There is a present challenge of assimilating data into systems that undergo abrupt change. Bimodal distributions arise naturally and defy Gaussian descriptions. What if data and model drastically diverge? Can understanding of underlying geometric structures, like MPEPs help in resolving this difficulty in a similar vein to Apte et al¹? My problems in this context will also be built around existing work. This is an ongoing research direction that will be advanced by more diverse collaborations in the field. My ultimate goal is to further develop new data assimilation schemes in low and high-order models to assess the validity of their predictions. I plan to design and mentor various student research projects in data assimilation. Undergraduates can build probabilistic models for data stratified by different factors (or groups). They can study whether the noisy data obtained can be assimilated to make predictions or simply for reanalysis purposes. A challenging project for a theoretically-focused student is to generalize our methods. This will require careful design of the corresponding optimization problem. Another interesting project would be to analyze whether tipping can be used to create positive effects. Social Tipping Interventions (STIs) can generate rapid change in behaviors, social norms, and structural reorganization within their functional domains.

External funding. Finally, I will craft appropriate funding applications, such as to the applied and/or computational math programs of DMS at the National Science Foundation (NSF), to allow for focused, interdisciplinary research projects that will involve student researchers and include travel for collaboration and conference participation. I have experience in writing grants since I already secured a NSF MPS Ascend Award (\$300,000) and am currently applying for additional NSF funding this academic cycle. Beyond NSF, I will pursue funding from the Department of Energy's Advanced Scientific Computing Research (ASCR) program for computational mathematics research and the Alfred P. Sloan Foundation's research fellowships for early-career researchers

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